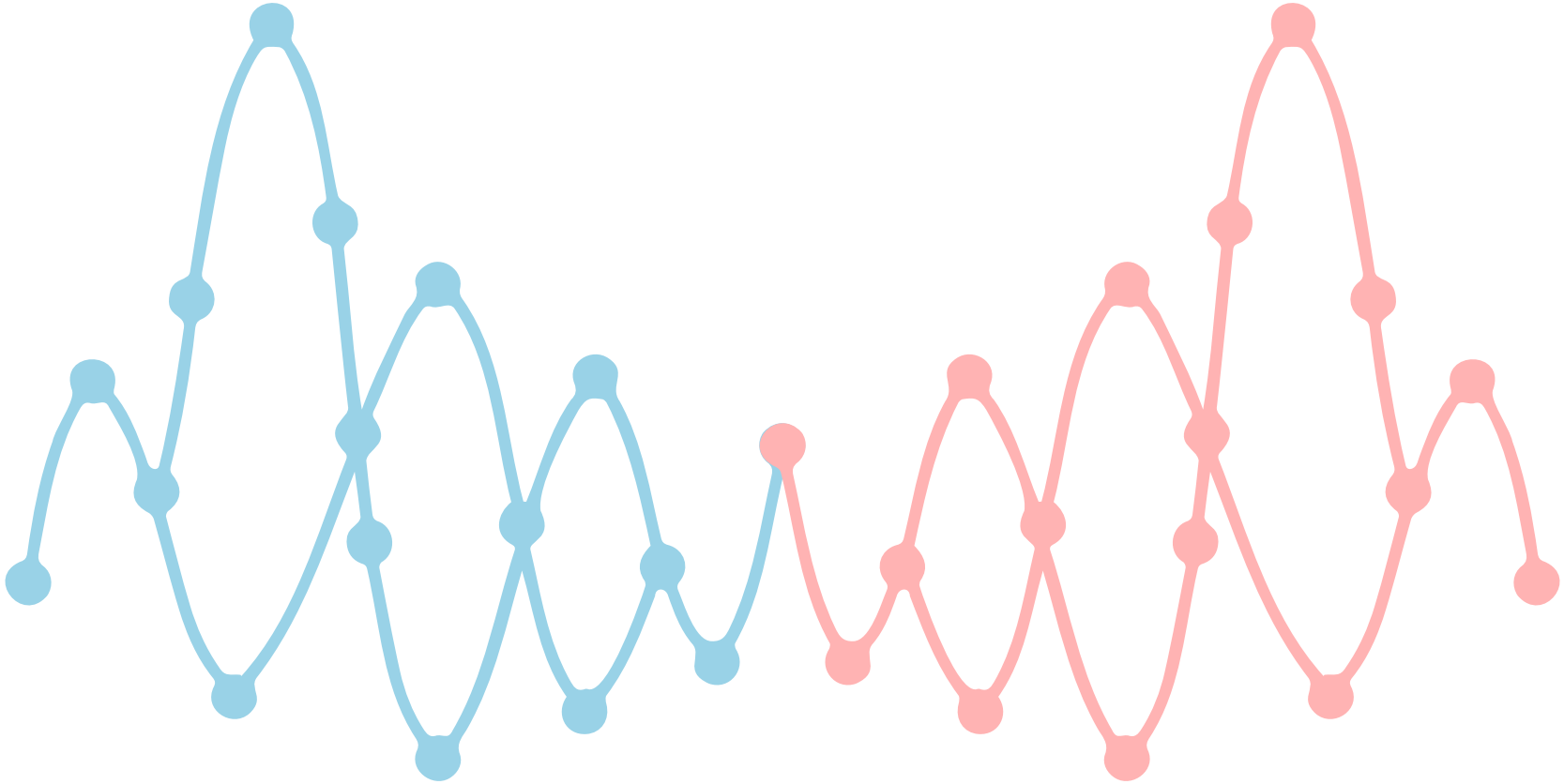
To Gossip or Not To Gossip

Ananya, Shrayansh, Yogesh



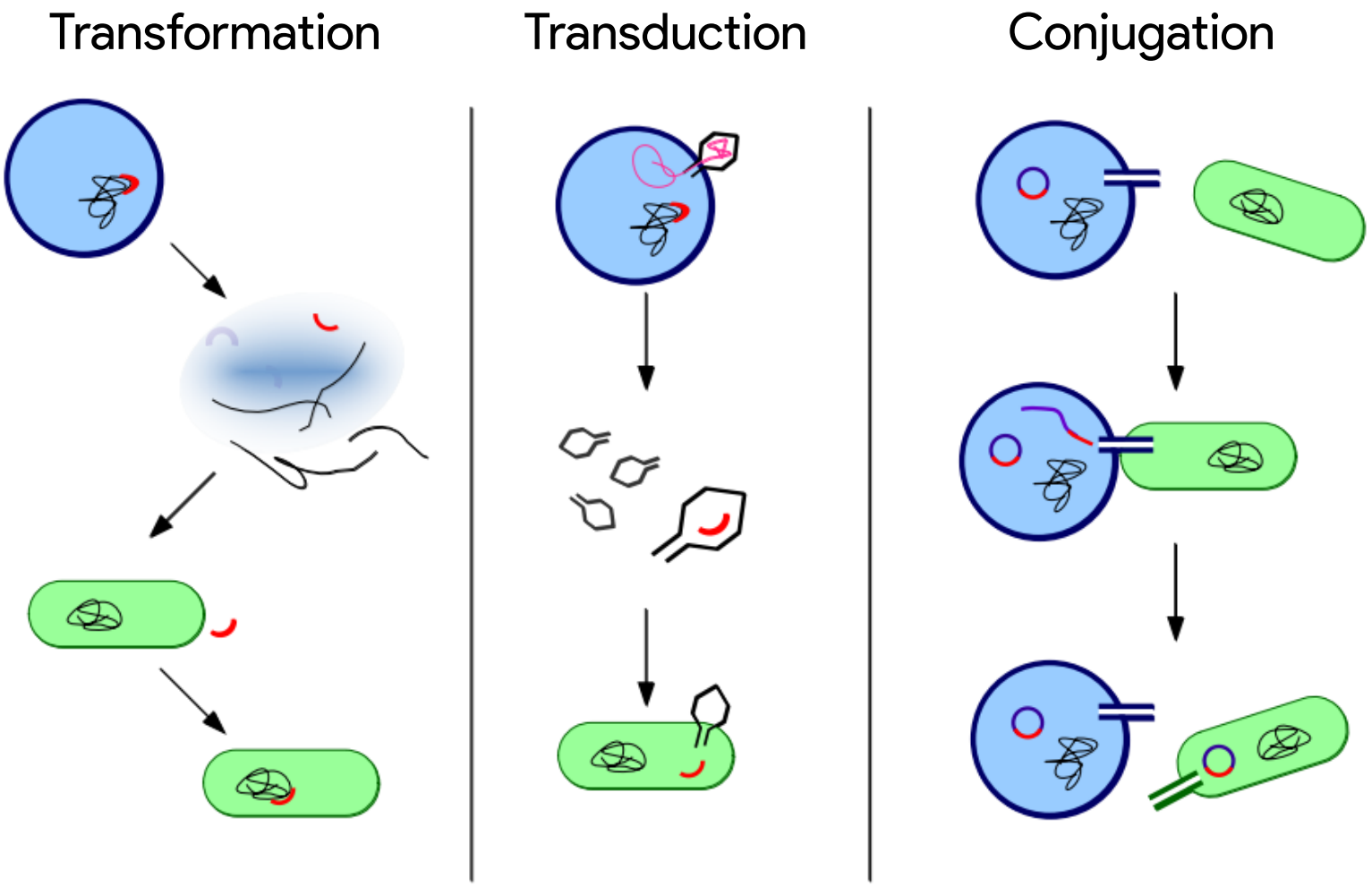
20/06/2026

Behavior, Evolution, Emergence 2026, IMSc

Gossip in Bacteria - Horizontal Gene Transfer

Transfer of genes from one bacteria to another bacteria without production of an offspring like in sexual reproduction

Horizontal Gene Transfer



Horizontal Gene Transfer

- Wide Spread. Almost all bacterial strains perform HGT

- 80% of all bacterial genome over history has been acquired through HGT

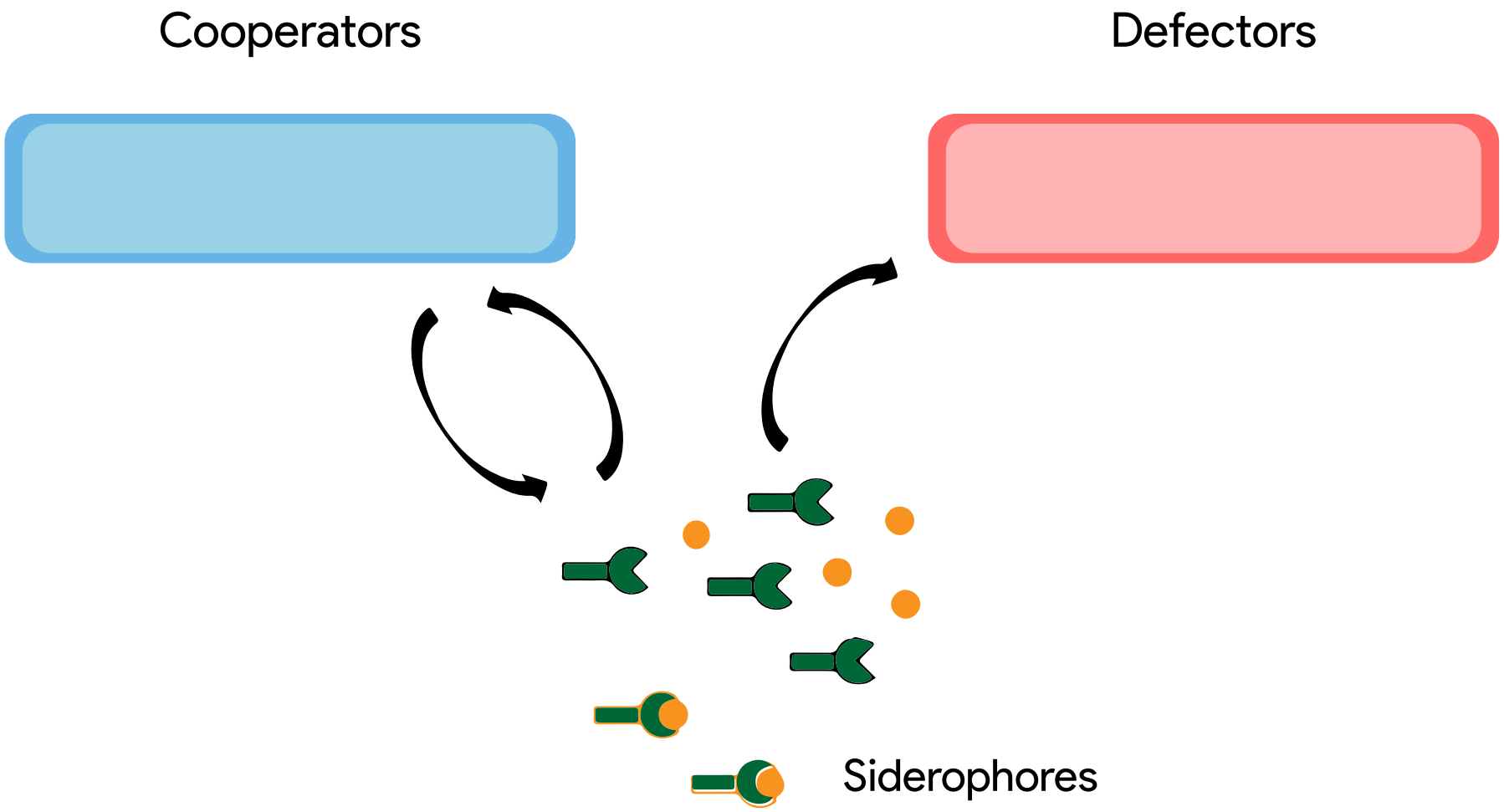
- Costly. Donor driven process - donor bacteria produces all the machinery for HGT

- Dono bacteria can give up beneficial genes increasing fitness of other strains

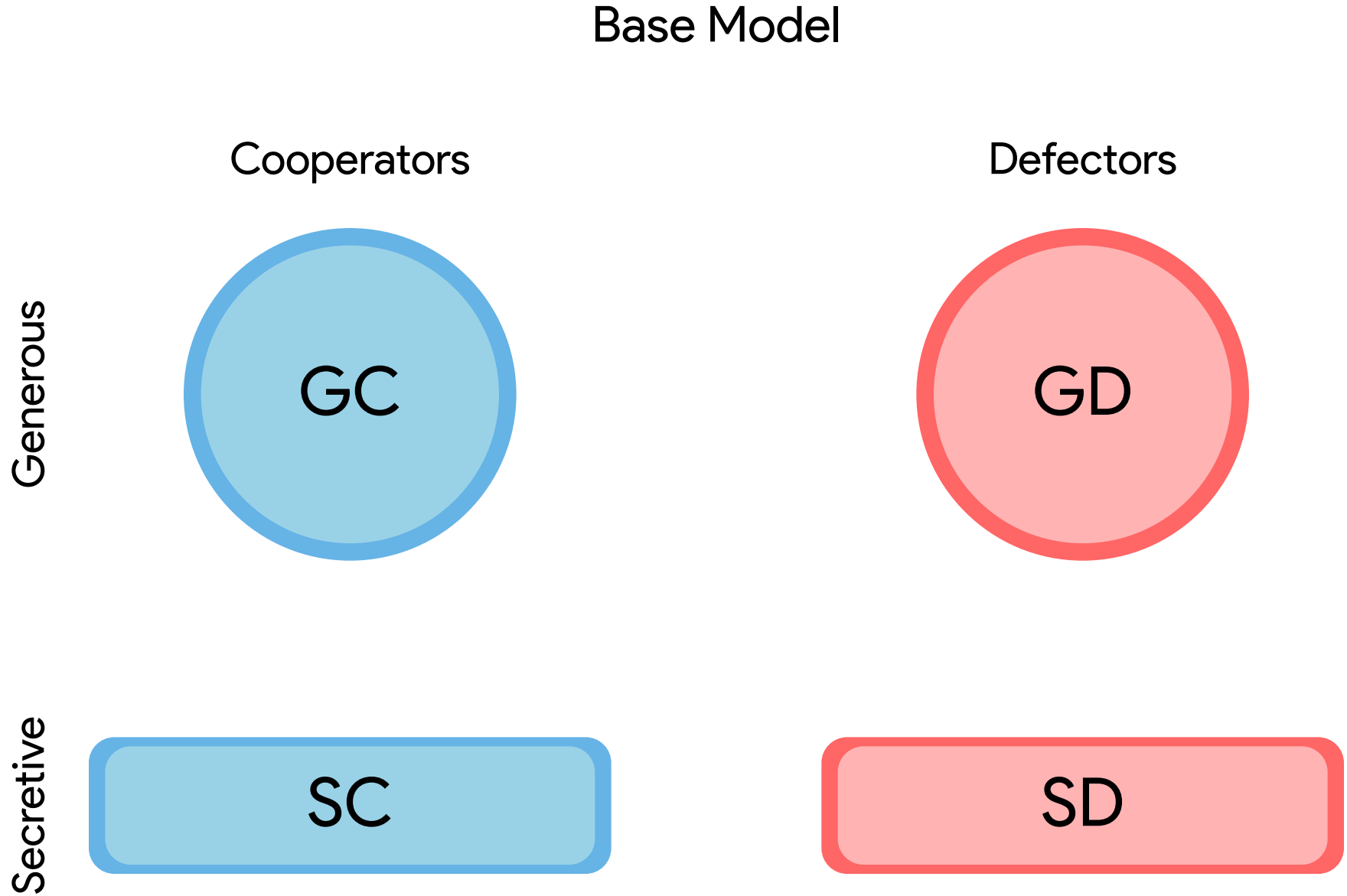
Why is HGT so widespread even though it is so costly?

Evolution of Horizontal Gene Transfer

Bacterial Public Goods Game



Evolution of Horizontal Gene Transfer in the context of a baterial public goods game


Base Model
Cooperators
Defectors
Generous
GC
GD
Secretive
SC
SD

Base Model

Conjugation Events

$$GC + SD \xrightarrow{\alpha} GC + SC$$
$$GC + GD \xrightarrow{\alpha} 2GC$$

Fitness of each types

$$f_{GC} = 1 - f_c$$
$$f_{GD} = 1$$
$$f_{SC} = 1 + f_s - f_c$$
$$f_{SD} = 1 + f_s$$

α - Rate of Conjugation

f_c - Cost of Cooperation

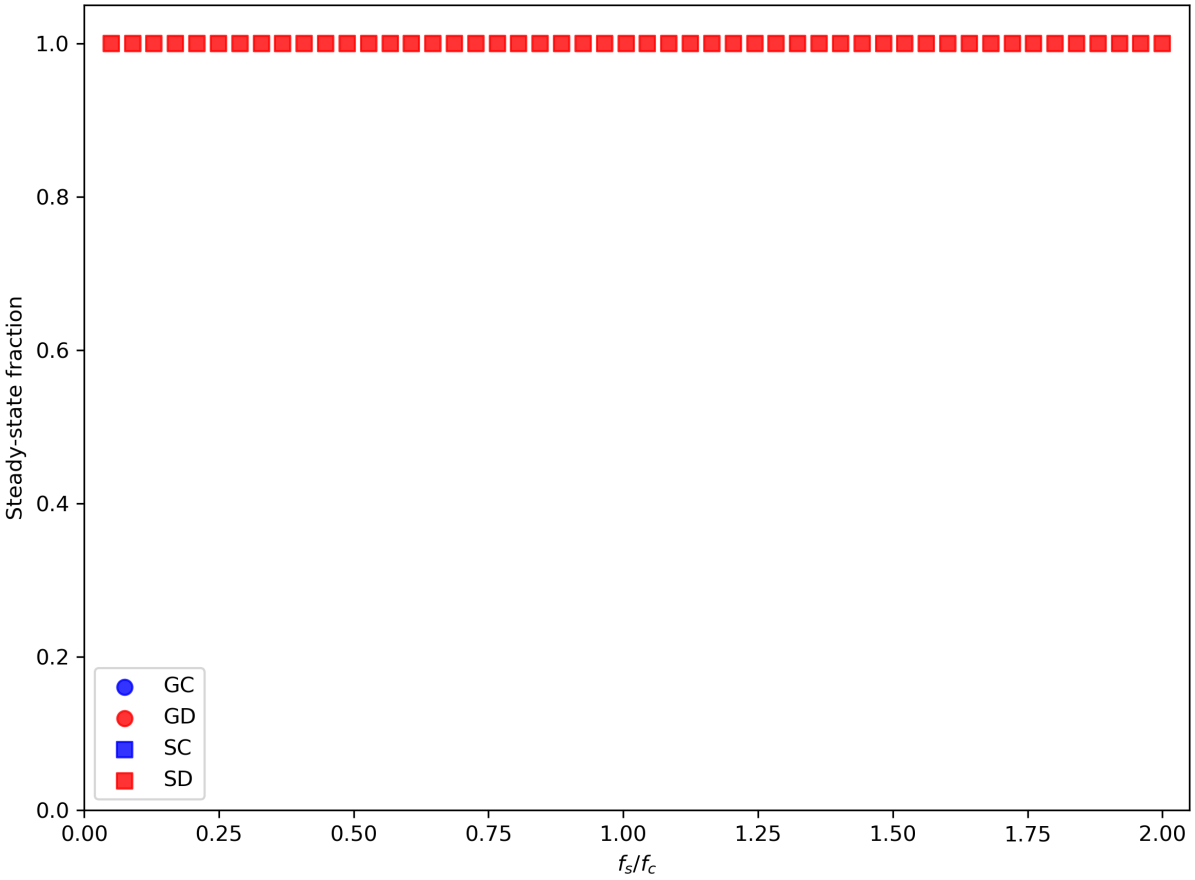
f_s - Cost of performing HGT

Base Model - Equation

$$\dot{GC} = (f_{GC} - \bar{f})GC + \alpha.GC.GD$$
$$\dot{GD} = (f_{GD} - \bar{f})GD - \alpha.GC.GD$$
$$\dot{SC} = (f_{SC} - \bar{f})SC + \alpha.GC.SD$$
$$\dot{SD} = (f_{SD} - \bar{f})SD - \alpha.GC.SD$$

$$\bar{f} = 1 - f_c(GC + SC) + f_s(SC + SD)$$

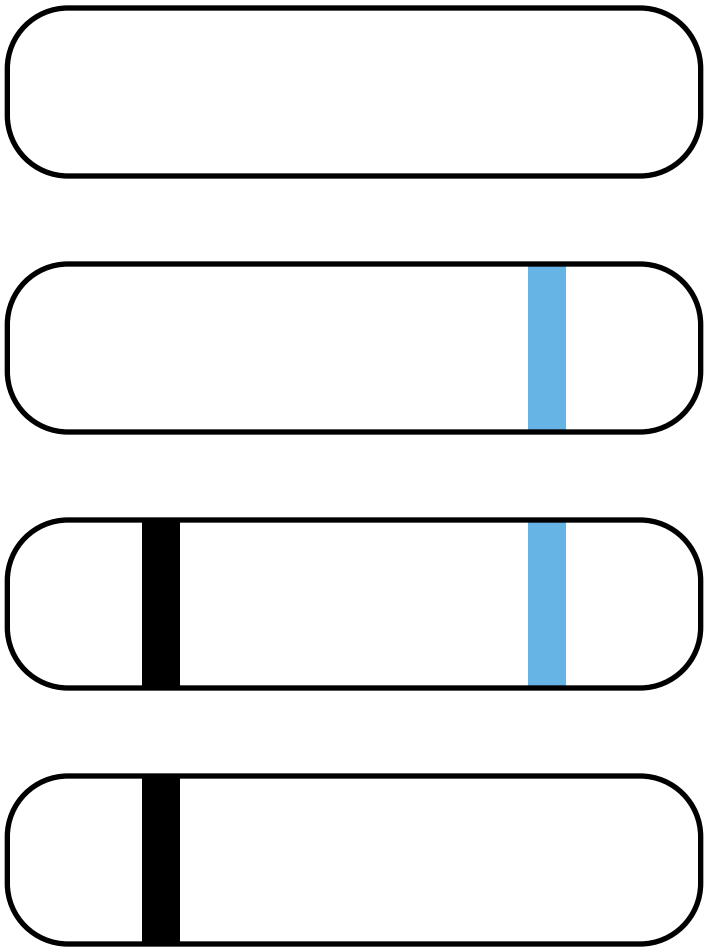
Base Model - Result



Stable State

SD

Independent Beneficial Plasmid Model



Plasmid

Independent Beneficial Plasmid Model

Conjugation Events





$GC_- + GD_- \rightarrow 2GC_-$
$GC_- + SD_+ \rightarrow GC_- + SC_+$

$GC_- + GD_- \rightarrow 2GC_-$
$GC_- + SD_+ \rightarrow GC_- + SC_+$



$GC_+ + GC_- \rightarrow 2GC_+$
$GC_+ + GD_+ \rightarrow 2GC_+$
$GC_+ + GD_- \rightarrow 2GC_+$

$GC_+ + SC_- \rightarrow SC_+ + GC_+$
$GC_+ + SD_+ \rightarrow SC_+ + GC_+$
$GC_+ + SD_- \rightarrow SC_+ + GC_+$



$GD_+ + GC_- \rightarrow GC_+ + GD_+$
$GD_+ + GD_- \rightarrow 2GD_+$

$GD_+ + SC_- \rightarrow SC_+ + GD_+$
$GD_+ + SD_- \rightarrow SD_+ + GD_+$

Independent Beneficial Plasmid Model

Fitness of each types

$$
\begin{aligned}
f_{GC_+} &= 1 - f_c + f_+ \\
f_{GC_-} &= 1 - f_c \\
f_{GD_+} &= 1 + f_+ \\
f_{GD_-} &= 1 \\
f_{SC_+} &= 1 + f_s - f_c + f_+ \\
f_{SC_-} &= 1 + f_s - f_c \\
f_{SD_+} &= 1 + f_s + f_+ \\
f_{SD_-} &= 1 + f_s
\end{aligned}
$$

α - Rate of Conjugation

f_c - Cost of Cooperation

f_s - Cost of performing HGT

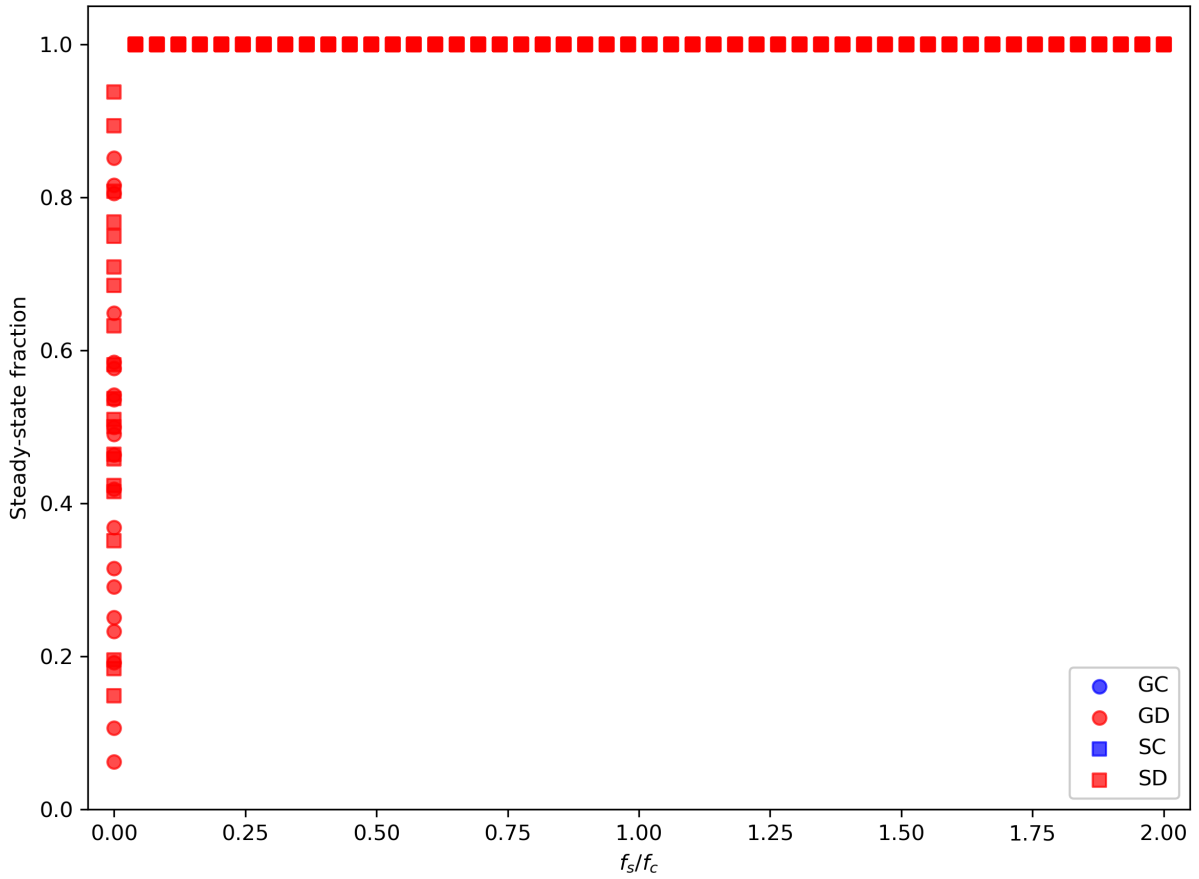
f_+- Gain from beneficial plasmid

Independent Beneficial Plasmid Model - Equation

$$\dot{GC_+} = (f_{GC_+} - \bar{f})GC_+ + \alpha(GC_+.GC_- + GC_+.GD_+ + GC_+.GD_- + 2.GC_-.GD_+)$$
$$\dot{GC_-} = (f_{GC_-} - \bar{f})GC_- + \alpha(GC_-.GD_- - GC_+.GC_- - GC_-.GD_+)$$
$$\dot{GD_+} = (f_{GD_+} - \bar{f})GD_+ + \alpha(GD_+.GD_- - GC_+.GD_+ - GC_-.GD_+)$$
$$\dot{GD_-} = (f_{GD_-} - \bar{f})GD_- + \alpha(-GC_+.GD_- - GC_-.GD_- - GD_+.GD_-)$$
$$\dot{SC_+} = (f_{SC_+} - \bar{f})SC_+ + \alpha(GC_+.SC_- + GC_+.SD_+ + GC_+.SD_- + GC_-.SD_+ + GD_+.SC_-)$$
$$\dot{SC_-} = (f_{SC_-} - \bar{f})SC_- + \alpha(GC_-.SD_- - GC_+.SC_- - GD_+.SC_-)$$
$$\dot{SD_+} = (f_{SD_+} - \bar{f})SD_+ + \alpha(GD_+.SD_- - GC_+.SD_+ - GC_-.SD_+)$$
$$\dot{SD_-} = (f_{SD_-} - \bar{f})SD_- + \alpha(-GD_+.SD_- - GC_+.SD_- - GC_-.SD_-)$$

$$\bar{f} = 1 - f_c(GC_+ + GC_- + SC_+ + SC_-) + f_s(SC_+ + SC_- + SD_+ + SD_-) + f_+(GC_+ + GD_+ + SC_+ + SD_+)$$

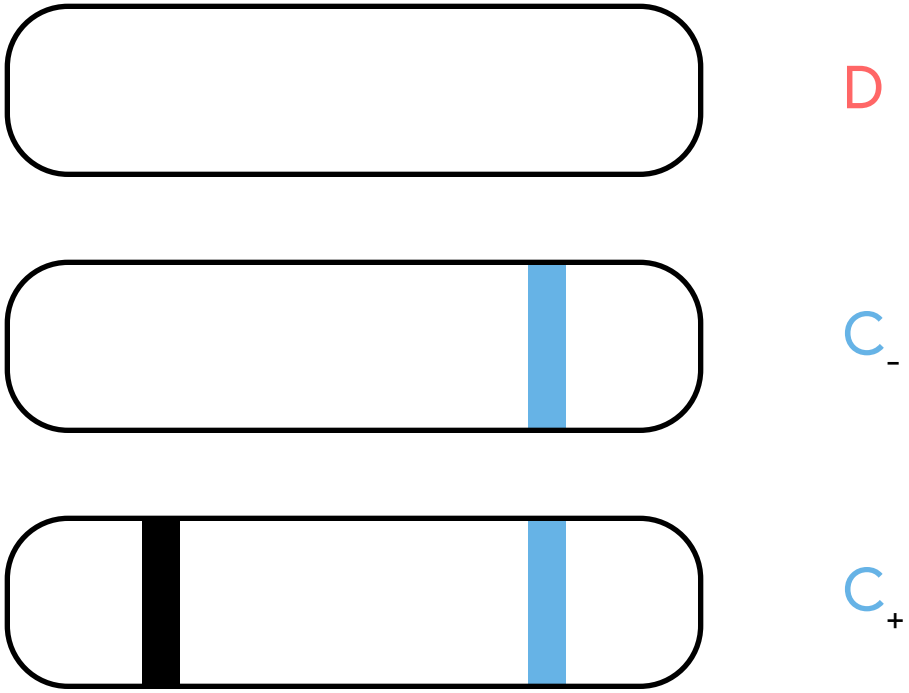
Independent Beneficial Plasmid Model - Result



Stable State

SD_+ + SD_-

Dependent Beneficial Plasmid Model



Plasmid

Dependent Beneficial Plasmid Model

Conjugation Events





$$GC_- + GD \rightarrow 2GC_-$$
$$GC_- + SD \rightarrow GC_- + SC_-$$



$$GC_+ + GD \rightarrow 2GC_+ \qquad GC_+ + SD \rightarrow SC_+ + GC_+$$
$$GC_+ + GC_- \rightarrow 2GC_+ \qquad GC_+ + SC_- \rightarrow SC_+ + GC_+$$

Dependent Beneficial Plasmid Model

Fitness of each types

$$
\begin{aligned}
f_{GC_+} &= 1 - f_c + f_+ \\
f_{GC_-} &= 1 - f_c \\
f_{GD} &= 1 \\
f_{SC_+} &= 1 + f_s - f_c + f_+ \\
f_{SC_-} &= 1 + f_s - f_c \\
f_{SD} &= 1 + f_s
\end{aligned}
$$

α - Rate of Conjugation

f_c - Cost of Cooperation

f_s - Cost of performing HGT

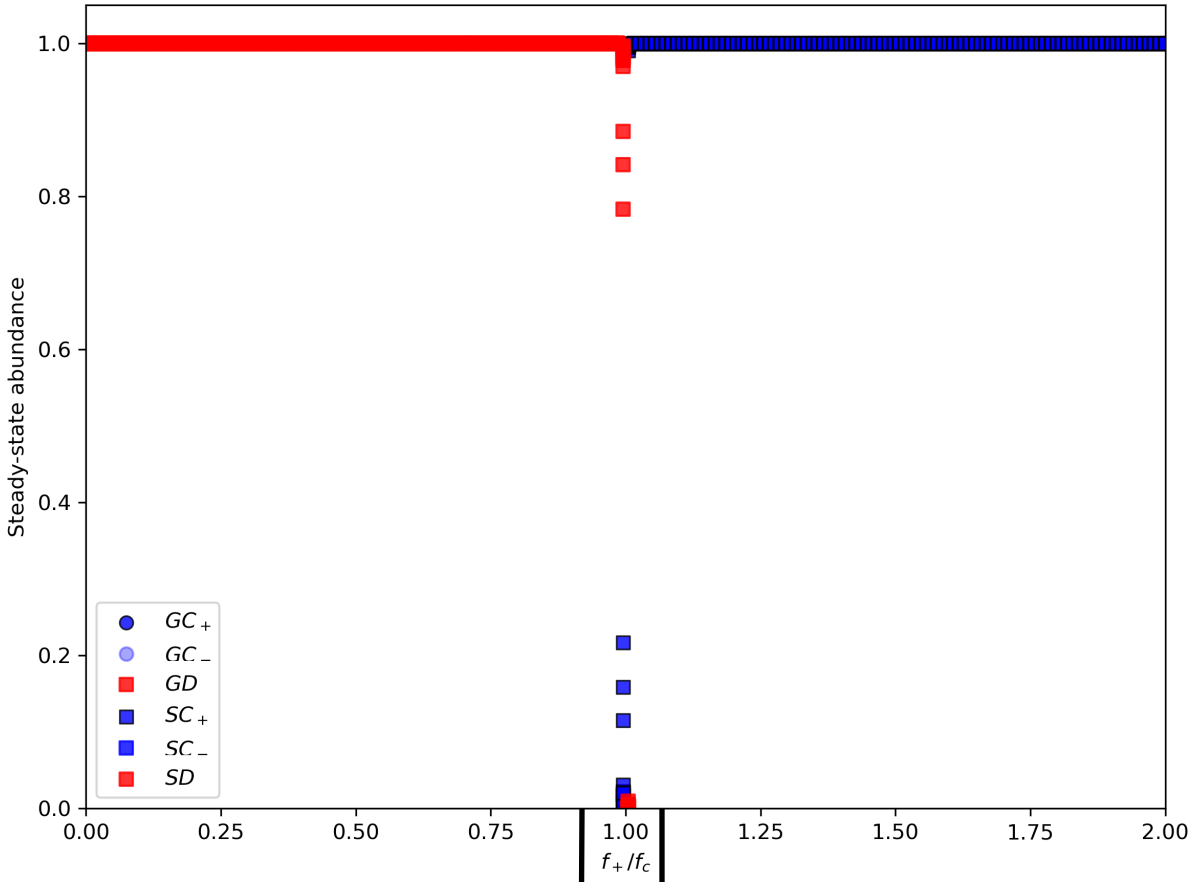
f_+- Gain from beneficial plasmid

Dependent Beneficial Plasmid Model - Equation

$$\dot{GC_+} = (f_{GC_+} - \bar{f})GC_+ + \alpha(GC_+.GD + GC_+.GC_-)$$
$$\dot{GC_-} = (f_{GC_-} - \bar{f})GC_- + \alpha(GC_-.GD + GC_-.SD - GC_+.GC_-)$$
$$\dot{GD} = (f_{GD} - \bar{f})GD - \alpha(GC_+.GD + GC_-.GD)$$
$$\dot{SC_+} = (f_{SC_+} - \bar{f})SC_+ + \alpha(GC_+.SD + SC_-.GC_+)$$
$$\dot{SC_-} = (f_{SC_-} - \bar{f})SC_- - \alpha.SC_-GC_+$$
$$\dot{SD} = (f_{SD} - \bar{f})SD - \alpha(GC_+.SD + GC_-.SD)$$

$$\bar{f} = 1 - f_c(GC_+ + GC_- + SC_+ + SC_-) + f_s(SC_+ + SC_- + SD) + f_+(GC_+ + SC_+)$$

Dependent Beneficial Plasmid Model - Result



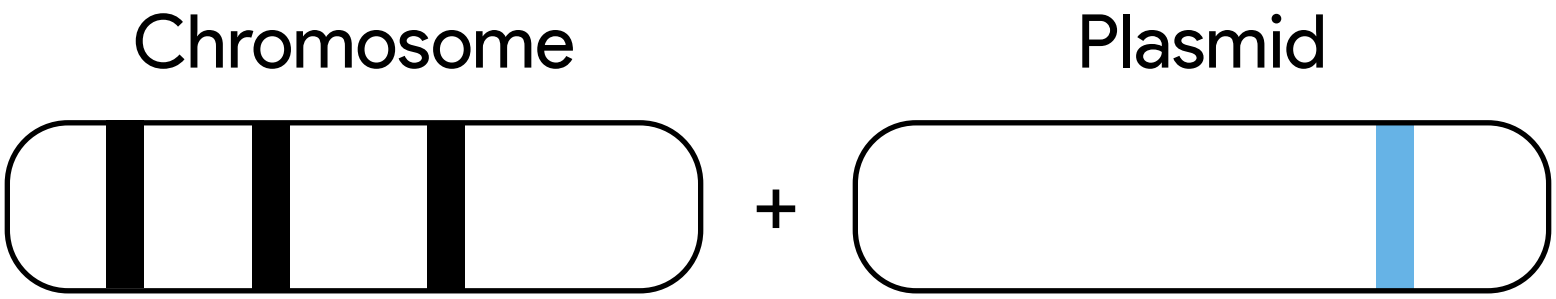
Stable State

SD

Coexistence

SC_+

Plasmid-Chromosome Synergy Model



Plasmid-Chromosome Synergy Model

Conjugation Events

$$GC + SD \xrightarrow{\alpha} GC + SC$$
$$GC + GD \xrightarrow{\alpha} 2GC$$

Fitness of each types

$$f_{GC} = 1 - f_c + f_{pc}$$
$$f_{GD} = 1$$
$$f_{SC} = 1 + f_s - f_c$$
$$f_{SD} = 1 + f_s$$

α - Rate of conjugation

f_s - Cost of conjugation

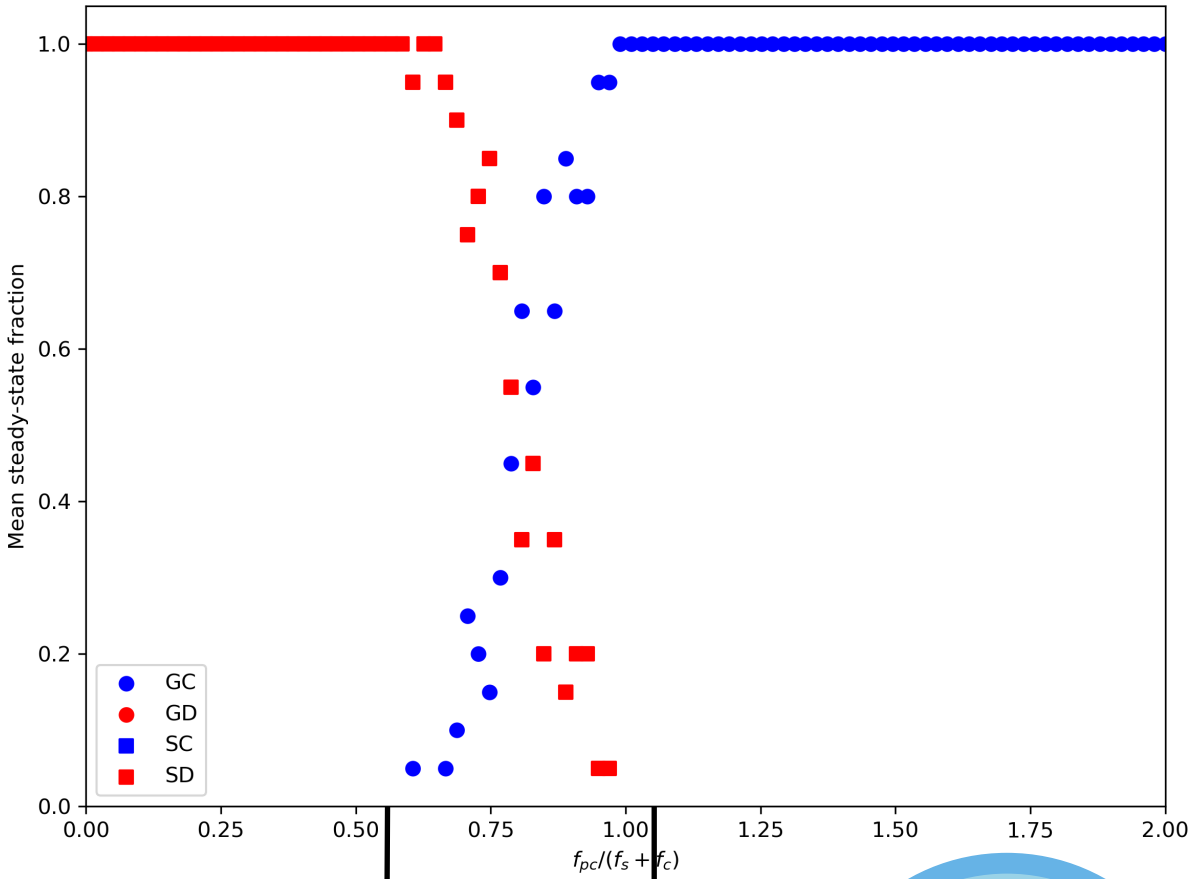
f_c - Cost of cooperation

f_{pc} - Gain from plasmid-chromosome synergy

Plasmid-Chromosome Synergy Model - Equation

$$\dot{GC} = (f_{GC} - \bar{f})GC + \alpha.GC.GD$$
$$\dot{GD} = (f_{GD} - \bar{f})GD - \alpha.GC.GD$$
$$\dot{SC} = (f_{SC} - \bar{f})SC + \alpha.GC.SD$$
$$\dot{SD} = (f_{SD} - \bar{f})SD - \alpha.GC.SD$$

Plasmid-Chromosome Synergy Model - Result



SD

Coexistence

GC

Stable State

Summary

- HGT can promote Cooperation in absence of classical cooperation enhancing mechanisms like space, kin transfer etc.

- Cooperation alone can't promote HGT except when it provides direct fitness benefits like in the synergy model

- The framework needs to analyzed further in presence of external cooperation enhancing mechanisms like space etc.

References

Alvaro San Millan, Karl Heilbron, R Craig MacLean, Positive epistasis between co-infecting plasmids promotes plasmid survival in bacterial populations, The ISME Journal, Volume 8, Issue 3, March 2014, Pages 601–612, https://doi.org/10.1038/ismej.2013.182

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Experimental evidence and cooperation https://www.pnas.org/doi/full/10.1073/pnas.1406840111

Public goods and cheating in microbes https://www.sciencedirect.com/science/article/pii/S0960982219302660

Genetic information transfer promotes cooperation in bacteria https://doi.org/10.1073/pnas.140684011

Bacterial cooperation through HGT https://www.cell.com/trends/ecology-evolution/fulltext/S0169-5347(21)00309-8?dgcid\x3draven_jbs_etoc_email

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Thank You